

#1.INTRODUCTION

Yulu Bike Rental Demand Analysis Using EDA and Hypothesis Testing

This analysis identifies weather and season as primary drivers of demand, with strong statistical evidence supporting their impact on bike rentals.

Yulu, a micro-mobility service provider, is experiencing a decline in revenues. Since revenue is directly dependent on the number of electric cycles rented (count), the objective of this analysis is to identify the key factors influencing demand.

The goals of study :-

- a) Understand how variables such as working day, season, weather, temperature, and user behavior affect bike rentals.
- b) Evaluate the statistical significance of these factors.
- c) Provide data-driven insights to improve demand and optimize business decisions.

#2.DATA UNDERSTANDING

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import IPython.display as display

# Load the bike_sharing.csv file into a pandas DataFrame
df_bike_sharing = pd.read_csv('bike_sharing.csv')

# Display the first 5 rows of the DataFrame
display.display(df_bike_sharing.head(5))

{"summary":{"\n  \"name\": \"display\",\n  \"rows\": 5,\n  \"fields\": [\n    {\n      \"column\": \"datetime\",\n      \"properties\": {\n        \"dtype\": \"object\",\n        \"num_unique_values\": 5,\n        \"samples\": [\n          \"2011-01-01 01:00:00\",\n          \"2011-01-01 04:00:00\",\n          \"2011-01-01 02:00:00\"\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    },\n    {\n      \"column\": \"season\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 0,\n        \"min\": 1,\n        \"max\": 1,\n        \"num_unique_values\": 1,\n        \"samples\": [\n          1\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    },\n    {\n      \"column\": \"holiday\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 0,\n        \"min\": 0,\n        \"max\": 0,\n        \"num_unique_values\": 1,\n        \"samples\": [\n          0\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    },\n    {\n      \"column\": \"workingday\",\n      \"properties\": {\n        \"dtype\": \"number\",
```


- b) Humidity ranges between 0% and 100%.
- c) Windspeed ranges up to ~56 km/h.
- d) Count demonstrates high variability with significant spread.

```
df_bike_sharing.shape
(10886, 12)
```

The dataset contains 10886 rows and 12 columns, representing hourly/daily bike rental data along with environmental and seasonal factors.

```
df_bike_sharing.info()
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10886 entries, 0 to 10885
Data columns (total 12 columns):
#   Column          Non-Null Count  Dtype
---  -
0   datetime        10886 non-null  object
1   season          10886 non-null  int64
2   holiday         10886 non-null  int64
3   workingday      10886 non-null  int64
4   weather         10886 non-null  int64
5   temp            10886 non-null  float64
6   atemp           10886 non-null  float64
7   humidity        10886 non-null  int64
8   windspeed       10886 non-null  float64
9   casual          10886 non-null  int64
10  registered      10886 non-null  int64
11  count           10886 non-null  int64
dtypes: float64(3), int64(8), object(1)
memory usage: 1020.7+ KB
```

The datetime is of object type which needs conversion. Also, season, holiday, workingday, weather are categorical columns but stored as integers. Therefore, Remaining variables are numerical

```
df_bike_sharing['datetime'] =
pd.to_datetime(df_bike_sharing['datetime'])

categorical_cols = ['season', 'holiday', 'workingday', 'weather']
for col in categorical_cols:
    df_bike_sharing[col] = df_bike_sharing[col].astype('category')
```

Converted relevant variables into categorical format to improve analysis and interpretation.

#3.Data Cleaning

```
df_bike_sharing.isnull().sum()
```

```
datetime    0
season      0
holiday     0
workingday  0
weather     0
temp        0
atemp       0
humidity    0
windspeed   0
casual      0
registered  0
count       0
dtype: int64
```

No missing values were found in the dataset, indicating clean and reliable data for analysis.

```
df_bike_sharing.duplicated().sum()
```

```
np.int64(0)
```

No duplicate records found as count is 0.

```
df_bike_sharing.describe()
```

```
{"summary": "{\n  \"name\": \"df_bike_sharing\",\n  \"rows\": 8,\n  \"fields\": [\n    {\n      \"column\": \"datetime\",\n      \"properties\": {\n        \"dtype\": \"date\",\n        \"min\": \"1970-01-01 00:00:00.000010886\",\n        \"max\": \"2012-12-19 23:00:00\",\n        \"num_unique_values\": 7,\n        \"samples\": [\n          \"10886\",\n          \"2011-12-27 05:56:22.399411968\",\n          \"2012-07-01 12:45:00\"\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    },\n    {\n      \"column\": \"temp\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 3842.2088126431295,\n        \"min\": 0.82,\n        \"max\": 10886.0,\n        \"num_unique_values\": 8,\n        \"samples\": [\n          20.23085981995223,\n          26.24,\n          10886.0\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    },\n    {\n      \"column\": \"atemp\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 3841.214609020895,\n        \"min\": 0.76,\n        \"max\": 10886.0,\n        \"num_unique_values\": 8,\n        \"samples\": [\n          23.655084052912,\n          31.06,\n          10886.0\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    },\n    {\n      \"column\": \"humidity\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 3830.3684503021896,\n        \"min\": 0.0,\n        \"max\": 10886.0,\n
```

```

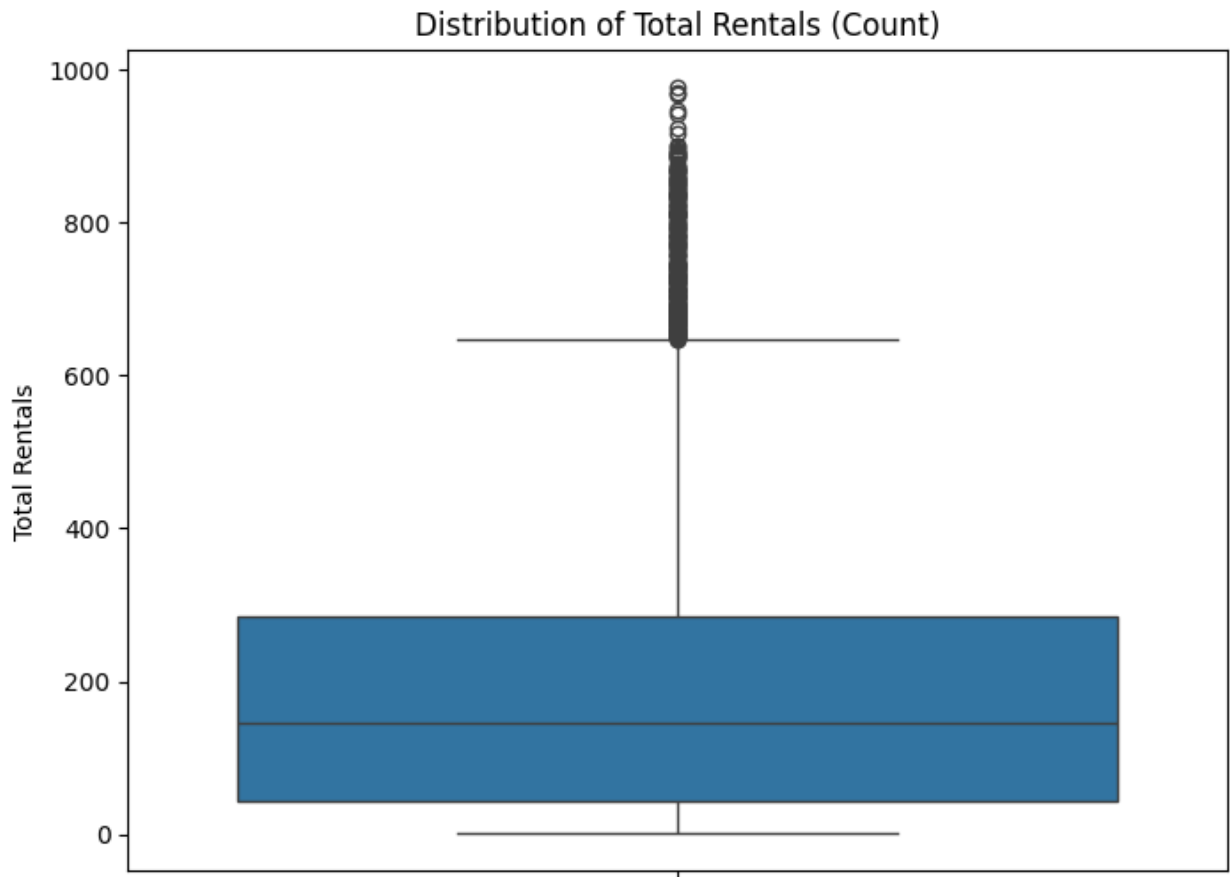
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\ "semantic_type\ ": \ "\",\n      \ "description\ ": \ "\"\n      }\n
n    },\n    {\n      \ "column\ ": \ "windspeed\ ",\n
\ "properties\ ": {\n      \ "dtype\ ": \ "number\ ",\n      \ "std\ ":
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\ "semantic_type\ ": \ "\",\n      \ "description\ ": \ "\"\n      }\n
n    },\n    {\n      \ "column\ ": \ "casual\ ",\n      \ "properties\ ":
{\n      \ "dtype\ ": \ "number\ ",\n      \ "std\ ":
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n      \ "num_unique_values\ ": 8,\n      \ "samples\ ": [\n
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\ "semantic_type\ ": \ "\",\n      \ "description\ ": \ "\"\n      }\n
n    },\n    {\n      \ "column\ ": \ "registered\ ",\n
\ "properties\ ": {\n      \ "dtype\ ": \ "number\ ",\n      \ "std\ ":
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\ "num_unique_values\ ": 8,\n      \ "samples\ ": [\n
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\ "semantic_type\ ": \ "\",\n      \ "description\ ": \ "\"\n      }\n
n    },\n    {\n      \ "column\ ": \ "count\ ",\n      \ "properties\ ": {\n
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\ "min\ ": 1.0,\n      \ "max\ ": 10886.0,\n
\ "num_unique_values\ ": 8,\n      \ "samples\ ": [\n
191.57413191254824,\n      284.0,\n      10886.0\n      ],\n
\ "semantic_type\ ": \ "\",\n      \ "description\ ": \ "\"\n      }\n
n    }\n  ]\n},"type":"dataframe"}

```

```

plt.figure(figsize=(8, 6))
sns.boxplot(y=df_bike_sharing['count'])
plt.title('Distribution of Total Rentals (Count)')
plt.ylabel('Total Rentals')
plt.show()

```



The outliers, indicates that the total number of rentals was significantly higher than the typical range.

```
for col in categorical_cols:
    print(col, df_bike_sharing[col].unique())

season [1, 2, 3, 4]
Categories (4, int64): [1, 2, 3, 4]
holiday [0, 1]
Categories (2, int64): [0, 1]
workingday [0, 1]
Categories (2, int64): [0, 1]
weather [1, 2, 3, 4]
Categories (4, int64): [1, 2, 3, 4]
```

Season has 4 categories (Spring, Summer, Fall, Winter) Weather has 4 categories ranging from clear to heavy rain Working day is binary (0,1)

Initial interpretation :-

- a)The dataset includes a mix of categorical and continuous variables influencing bike demand.
- b)No missing or duplicate values ensure high data quality.

c) Demand (count) varies significantly, indicating influence from multiple factors.

d) Environmental variables like temperature, humidity, and weather are expected to play a key role.

#3.EDA(Exploratory DATA ANALYSIS)

Univariate Analysis

Continuous variables :- temp, atemp, humidity, windspeed, count

```
import seaborn as sns

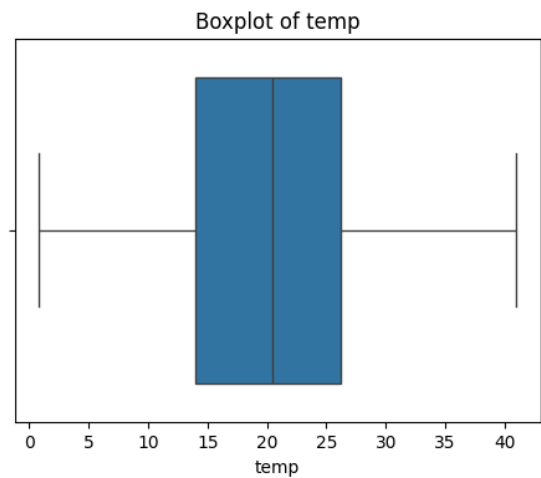
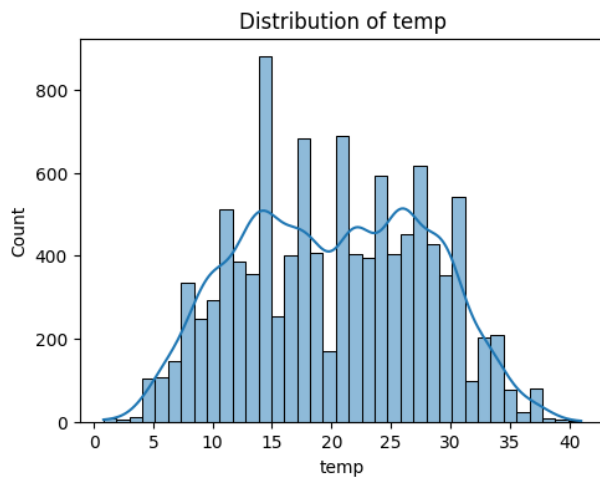
cont_cols = ['temp', 'atemp', 'humidity', 'windspeed', 'count']

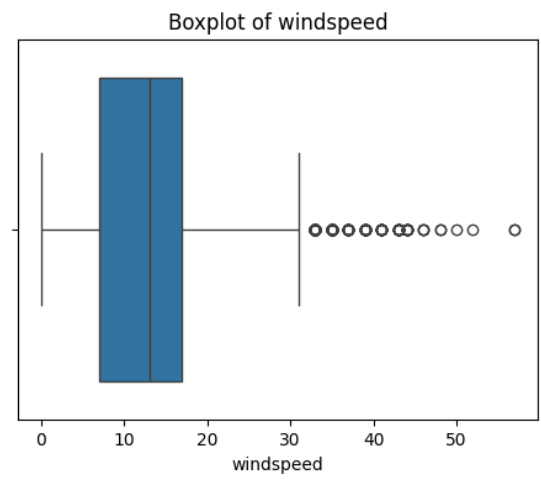
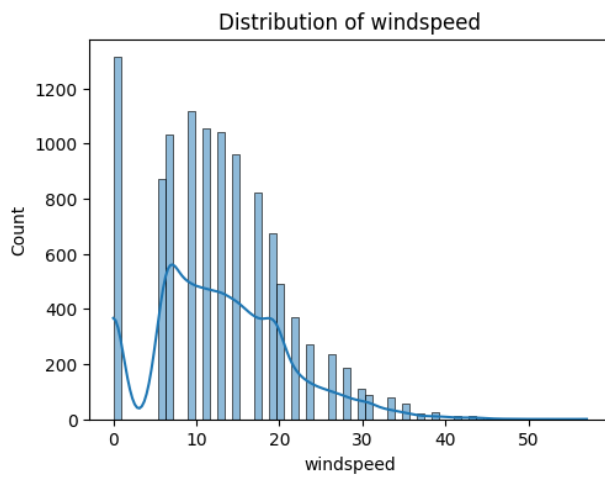
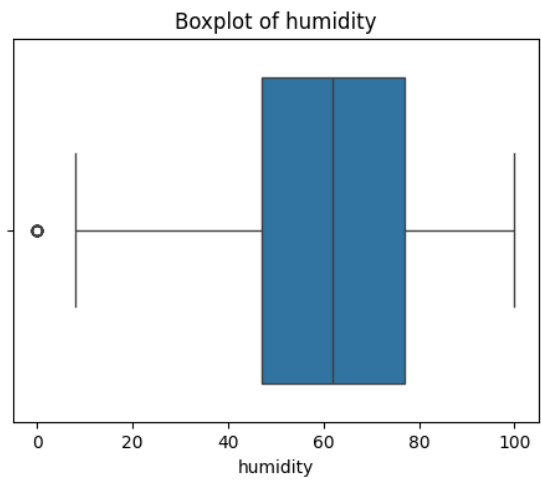
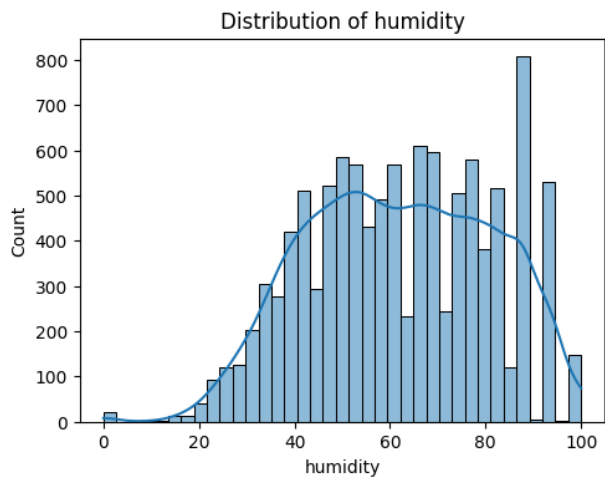
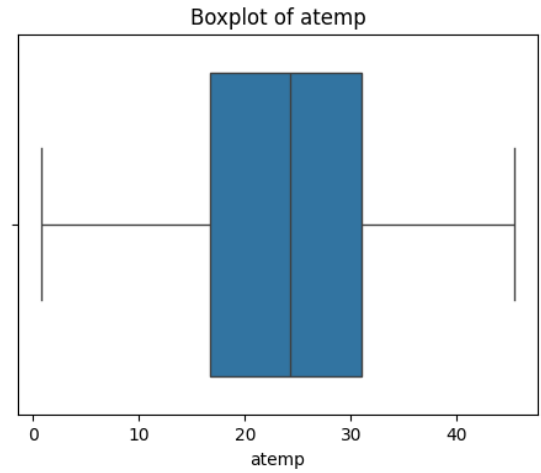
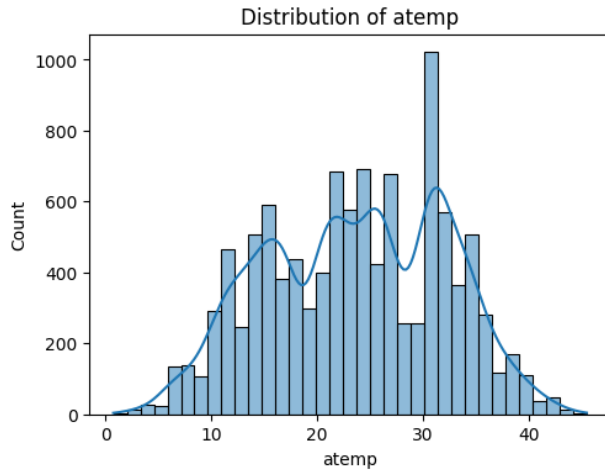
for col in cont_cols:
    plt.figure(figsize=(12,4))

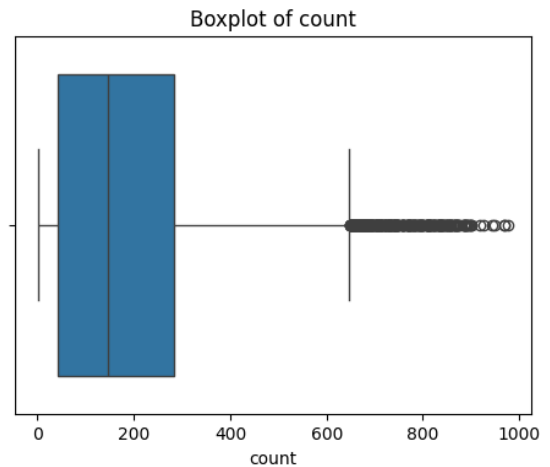
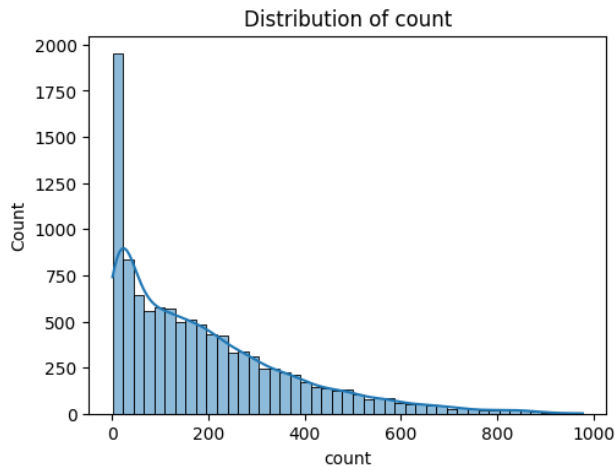
    plt.subplot(1,2,1)
    sns.histplot(df_bike_sharing[col], kde=True)
    plt.title(f'Distribution of {col}')

    plt.subplot(1,2,2)
    sns.boxplot(x=df_bike_sharing[col])
    plt.title(f'Boxplot of {col}')

    plt.show()
```







Key points:-

a)temp & atemp with temperature and feeling temperature indicates near-normal distributions and strong similarity, indicating a high correlation and consistent environmental measurement.

b)Humidity is slightly right-skewed with a wide spread, showing moderate variation which may influence rider comfort and demand.

c)Windspeed is highly right-skewed with many low values and few extreme highs (outliers), suggesting most days have low wind but a few extreme conditions.

d)The target variable count is right-skewed and exhibits high variance, indicating that low rental days are more frequent than high-demand days.

Conclusion:-

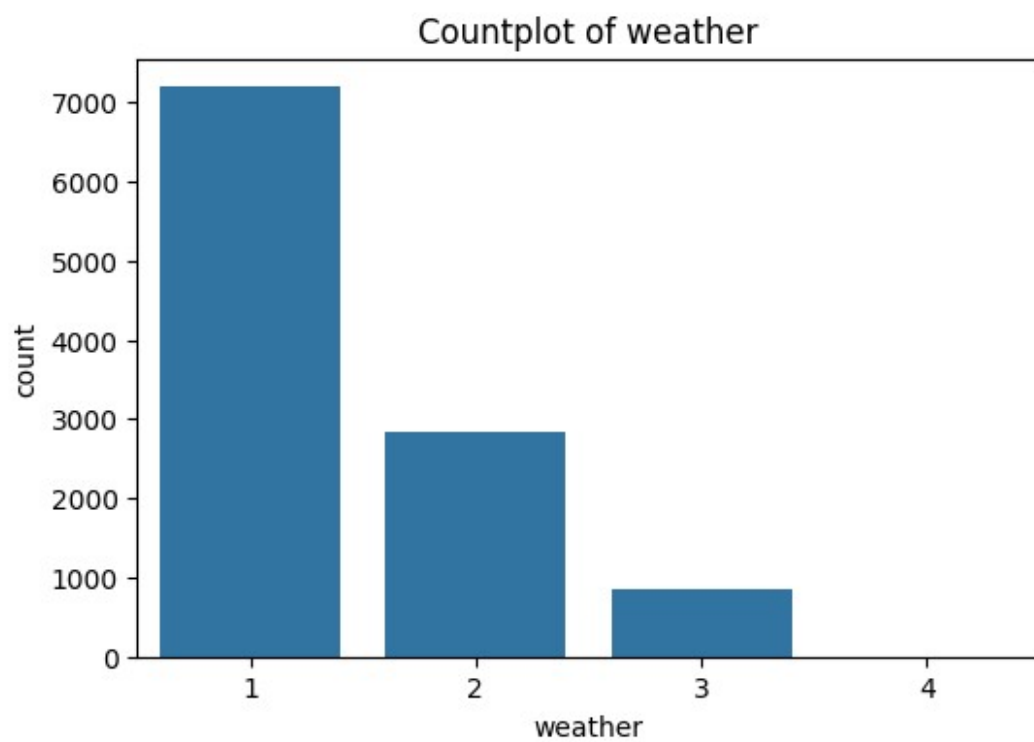
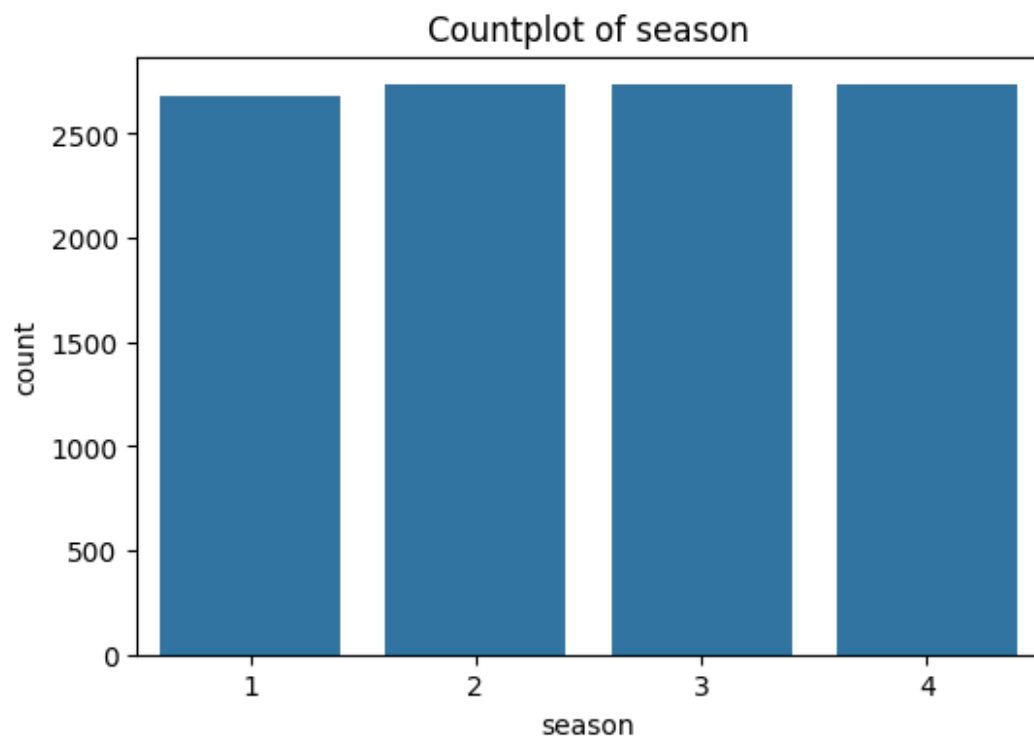
The high variability in demand across these factors strongly suggests the presence of significant influencing factors such as weather and seasonality.

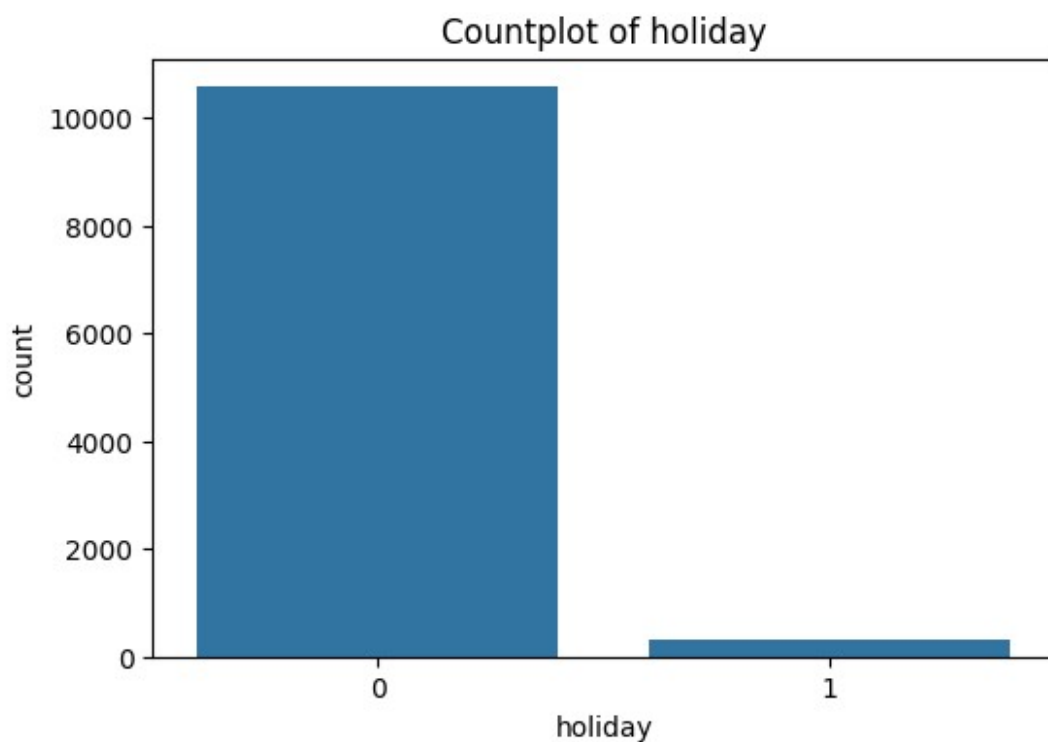
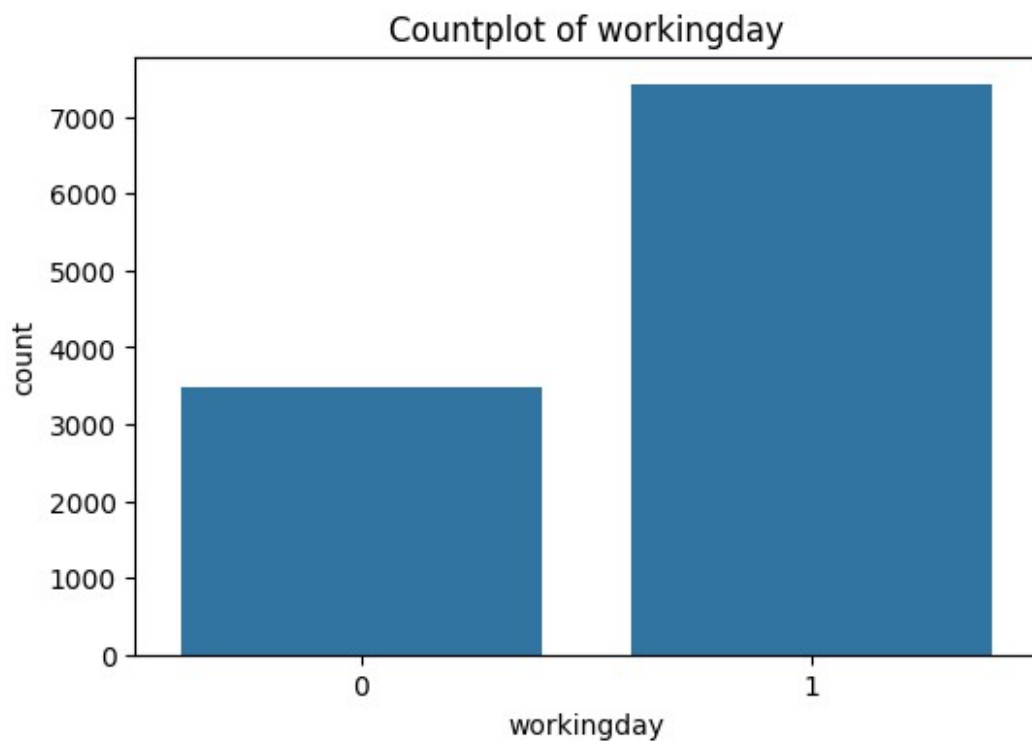
Categorical Variables:-

season, weather, workingday, holiday

```
cat_cols = ['season', 'weather', 'workingday', 'holiday']

for col in cat_cols:
    plt.figure(figsize=(6,4))
    sns.countplot(x=df_bike_sharing[col])
    plt.title(f'Countplot of {col}')
    plt.show()
```





Key points:-

a) Data is fairly distributed across all seasons, ensuring balanced seasonal analysis.

b) Majority of data fall under clear or moderate weather conditions, with very few extreme weather instances.

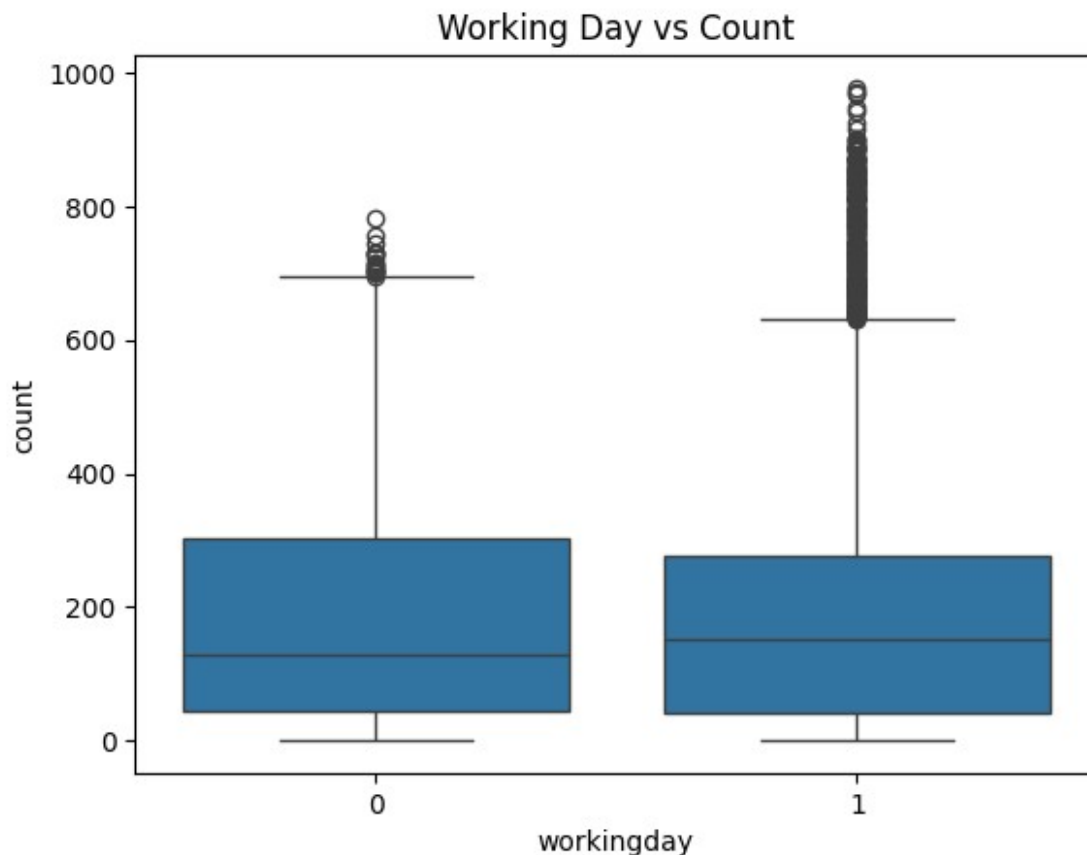
c) Most observations correspond to working days, indicating higher representation of weekday usage.

d) Holidays are significantly fewer compared to non-holidays.

Bivariate analysis

The aim is to Understand relationship between independent variables and demand (count).

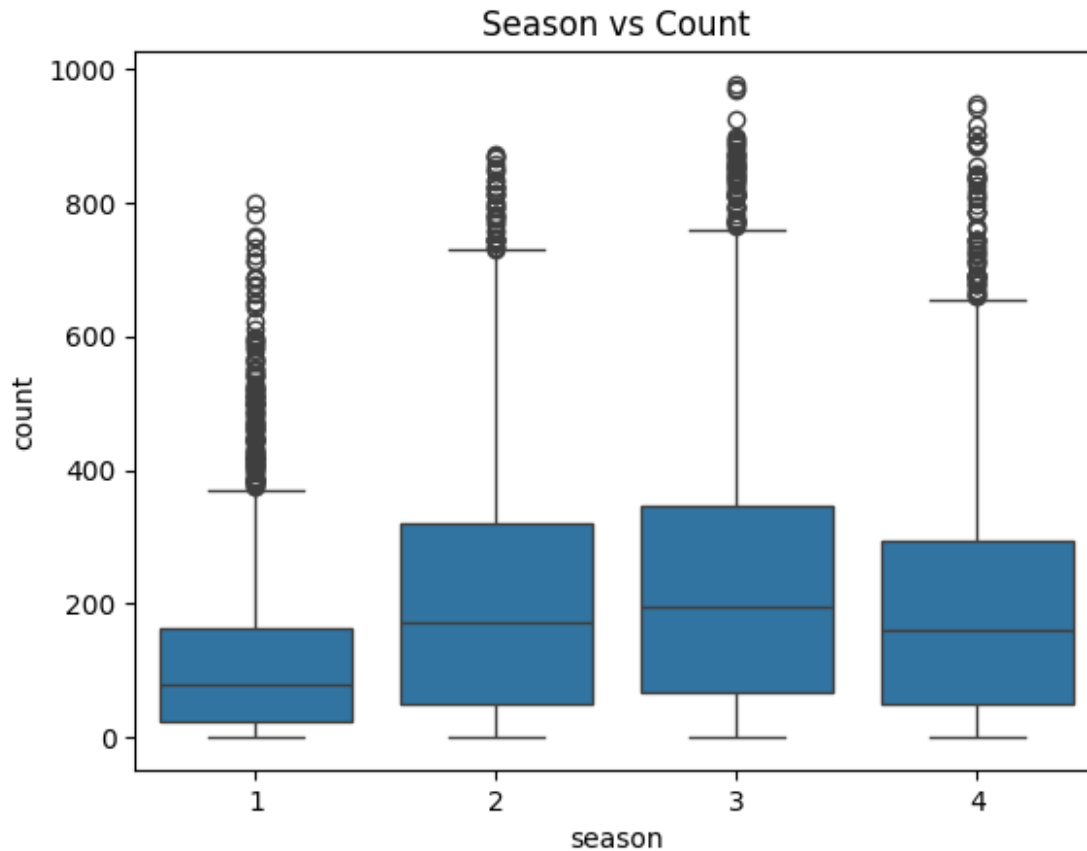
```
sns.boxplot(x='workingday', y='count', data=df_bike_sharing)
plt.title('Working Day vs Count')
plt.show()
```



Key insight :-

Rentals on working and non-working days indicates little variation, this difference in rentals between working and non-working days is not statistically significant, indicating that demand is not strongly driven by commuting patterns alone, suggesting limited impact.

```
sns.boxplot(x='season', y='count', data=df_bike_sharing)
plt.title('Season vs Count')
plt.show()
```

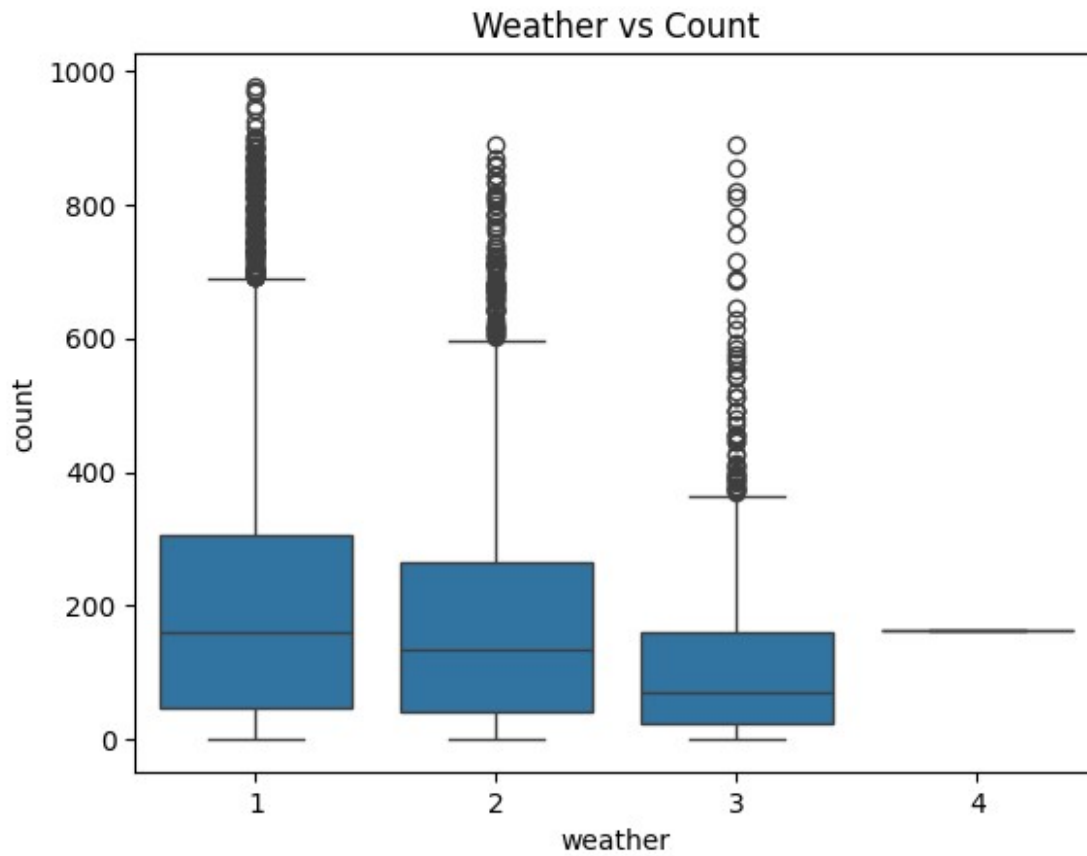


Key insights :-

a) Rentals vary significantly across seasons, with higher demand observed in summer and fall compared to winter.

b) This indicates strong seasonality in bike usage patterns.

```
sns.boxplot(x='weather', y='count', data=df_bike_sharing)
plt.title('Weather vs Count')
plt.show()
```

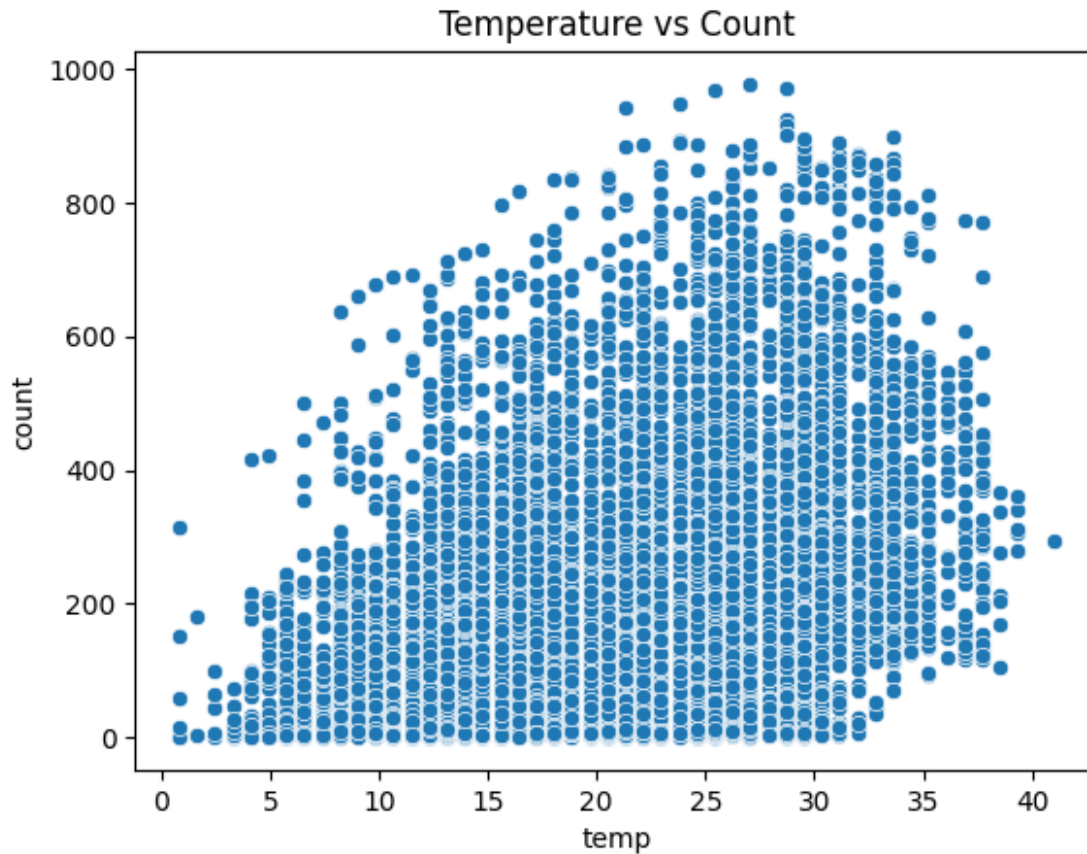


Key insights:-

a) Demand decreases sharply as weather conditions worsen, with the lowest rentals observed during heavy rain or extreme weather.

b) Weather appears to be one of the most critical factors influencing demand.

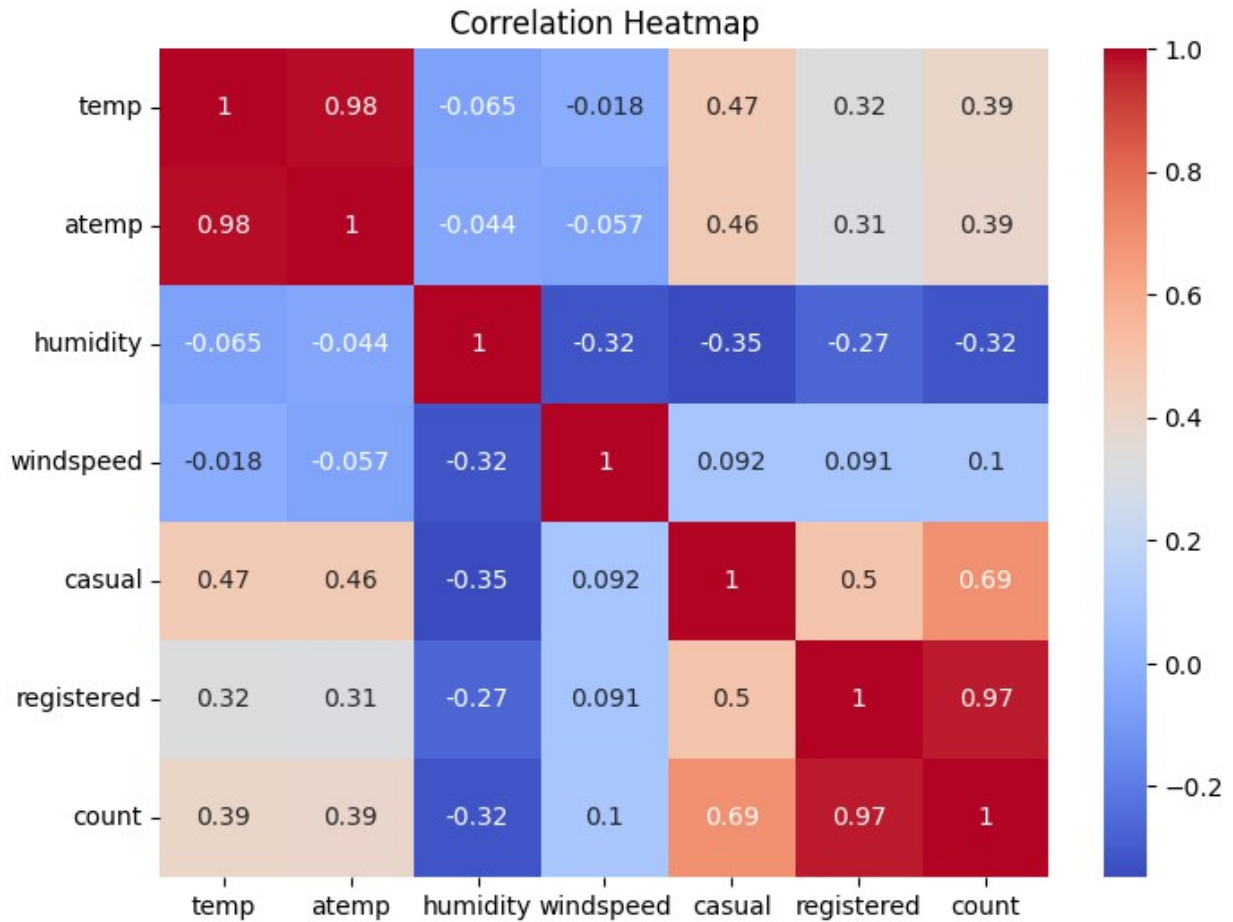
```
sns.scatterplot(x='temp', y='count', data=df_bike_sharing)
plt.title('Temperature vs Count')
plt.show()
```



key insight :-

A positive relationship is observed between temperature and demand, indicating increased rentals in moderate to warm conditions.

```
plt.figure(figsize=(8,6))
sns.heatmap(df_bike_sharing.corr(numeric_only=True), annot=True,
cmap='coolwarm')
plt.title('Correlation Heatmap')
plt.show()
```



Key insights:-

- a) Temp & atemp are highly correlated.
- b) Count moderately correlated with temp.
- c) Weak correlation with windspeed.

Illustrate the insights based on EDA

Comments on range and outliers

a) Temperature(temp, atemp)

*Range demonstrates moderate to high variation across seasons.

*No extreme outliers observed.

*temp and atemp move together consistent measurement.

insight :-

Temperature and feeling temperature demonstrates a stable range without extreme outliers, indicating reliable environmental conditions across conclusion.

b)humidity

- *Wide range from low to very high.

- *Some extreme values present.

Observation :-

Humidity demonstrates high variability with some extreme values, which may negatively impact rider comfort and reduce demand.

c)Windspeed:-

- *Wide range from low to extremely high.

- *Some extreme values present.

insight :-

Windspeed depicts high diversity with several high data points, indicating occasional strong wind conditions that may impact bike usage.

d)Count (Target Variable)

- *Highly wide range.

- *High standard deviation.

- *Few extreme high-demand points.

Business implication:-

The rental count exhibits high variability with a wide range, indicating fluctuating demand influenced by multiple external factors.

Distribution of variables:-

a)Continuous Variables

- *temp, atemp indicates approximately normal distribution graphically.

- *humidity demonstrates slightly skewed and windspeed represents heavily right-skewed in sub-plot representation.

- *count as a dependent variable visually represents right-skewed.

Business implication:-

- *Most environmental variables follow a reasonable distribution, while windspeed and count are right-skewed, indicating that extreme conditions and high demand occur less frequently.

- *The skewness in count suggests that peak demand days are limited, making demand prediction challenging.

Categorical variables

a)Season

*It represents that data is evenly distributed.

b)Weather

*It shows its domination in clearly in various conditions.

c)Workingday

*There are more working days then non-working days.

d)Holiday

*This variable shows less number of holidays.

Business implication:-

The dataset is well-balanced across seasons but skewed toward clear weather and working days, which may influence overall demand patterns.

Relationship Between Variables (Bivariate Insights)

a)Working day vs Count

The difference in bike rentals between working and non-working days is relatively small, indicating that working day status has a limited effect on demand.

Business insight:-

This suggests that bike usage is not strictly dependent on commuting patterns alone.

b)Season vs Count

Significant variation in rentals is observed across seasons, with higher demand during summer and fall, and lower demand during winter.

Business insight:-

Seasonal trends strongly influence demand, likely due to weather comfort and outdoor activity levels.

Weather vs Count

Rentals decrease sharply as weather conditions worsen, with the lowest demand observed during heavy rain or extreme weather.

Business insights:-

Weather is a critical determinant of bike usage, directly impacting user willingness to ride.

Temperature vs Count

A positive relationship exists between temperature and bike rentals, with demand increasing in moderate to warm conditions.

Business insight:-

Extremely high or low temperatures may reduce demand, indicating an optimal temperature range for usage.

Correlation analysis:-

Temperature and feeling temperature are highly correlated, while demand shows moderate positive correlation with temperature and weak correlation with windspeed.

Final EDA summary:-

- *Demand for electric cycles is highly influenced by weather and seasonal factors.

- *Temperature plays a positive role, encouraging higher rentals.

- *Extreme weather conditions significantly reduce demand.

- *Working day impact is comparatively weak, indicating mixed usage patterns (commute + leisure).

- *Presence of skewness and outliers suggests variability in demand, requiring statistical validation.

#5.HYPOTHESIS TESTING

2.Hypothesis testing

To check whether working day status impacts bike rentals (count)

Null Hypothesis (H_0):

There is no significant difference in the mean number of rentals between working days and non-working days.

Alternate Hypothesis (H_1):

There is a significant difference in the mean number of rentals between working days and non-working days.

```
working_day = df_bike_sharing[df_bike_sharing['workingday'] == 1]
['count']
non_working_day = df_bike_sharing[df_bike_sharing['workingday'] == 0]
['count']
```

Check normality (Shapiro Test)

```
from scipy.stats import shapiro

print(shapiro(working_day.sample(500)))
print(shapiro(non_working_day.sample(500)))
```

```
ShapiroResult(statistic=np.float64(0.868859428436609),  
pvalue=np.float64(4.399574801950338e-20))  
ShapiroResult(statistic=np.float64(0.8867293918225428),  
pvalue=np.float64(1.0600598290328493e-18))
```

Shapiro test may indicate non-normality due to large sample size. However, due to the Central Limit Theorem, T-test remains valid.

Equal variance (Levene test)

```
from scipy.stats import levene  
  
levene(working_day, non_working_day)  
  
LeveneResult(statistic=np.float64(0.004972848886504472),  
pvalue=np.float64(0.9437823280916695))
```

If p-value > 0.05 then equal variance assumed.

If p-value < 0.05 then use Welch's t-test.

```
from scipy.stats import ttest_ind  
  
ttest_ind(working_day, non_working_day, equal_var=False)  
  
TtestResult(statistic=np.float64(1.2362580418223226),  
pvalue=np.float64(0.21640312280695098),  
df=np.float64(7176.288413844308))
```

Set significant level (α) = 0.05

Decision :-

p-value = 0.21640312280695098(t-test)

If p-value < 0.05 then Reject H_0 .

If p-value $\geq$ 0.05 the Fail to reject H_0 .

Hence we Fail to Reject H_0 .

Mean comparison

```
working_day.mean(), non_working_day.mean()  
  
(np.float64(193.01187263896384), np.float64(188.50662061024755))
```

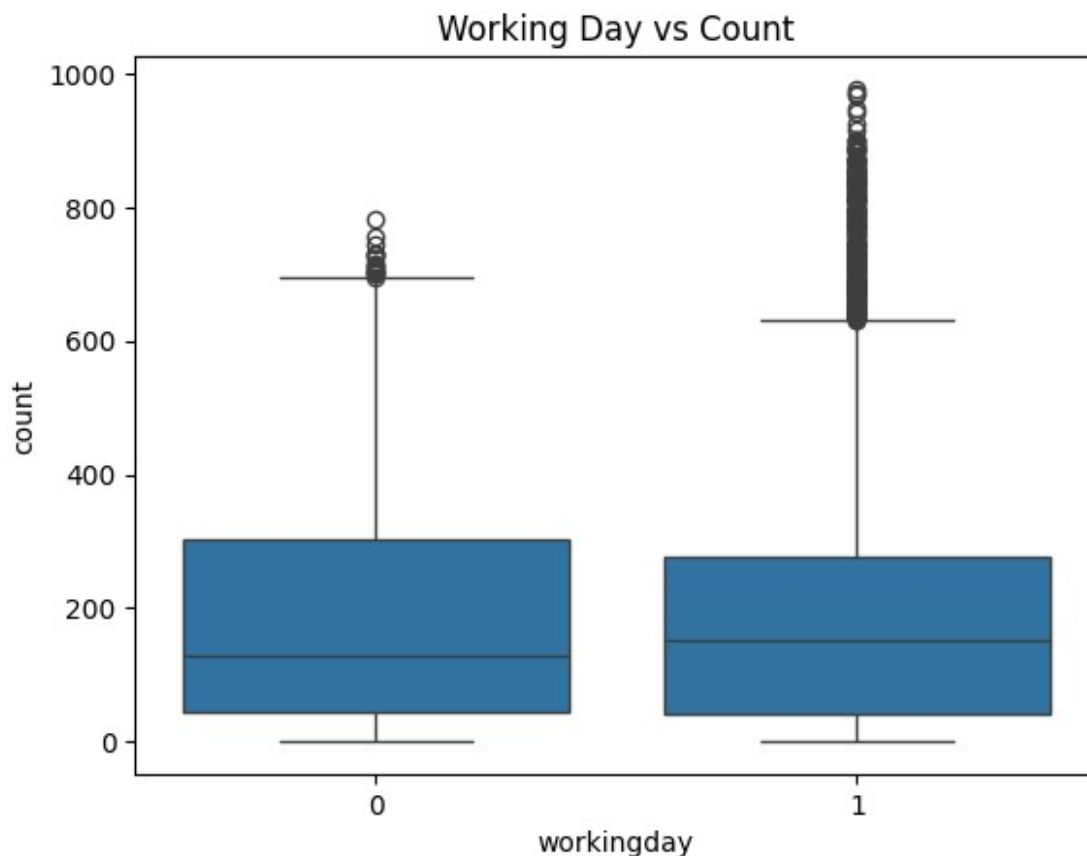
Observations:-

*The average rentals on working days are slightly higher than non-working days, but the difference is not statistically significant.

*The difference between means is small, indicating a negligible practical impact of working day on demand.

Boxplot:-

```
sns.boxplot(x='workingday', y='count', data=df_bike_sharing)
plt.title('Working Day vs Count')
plt.show()
```



Insights:-

*As per previous reference, The data depicts difference between working and non-working days, supporting the statistical result.

*The boxplot demonstrates overlapping distributions, reinforcing that there is no meaningful separation between the two groups.

Overall Assumption:-

The Shapiro-Wilk test indicates non-normality ($p < 0.05$). However, due to the large sample size, the Central Limit Theorem ensures robustness of the T-test. Levene's test confirms equal variance ($p > 0.05$), validating the use of standard independent T-test. Although normality assumption is violated, the large sample size ensures robustness of the test results.

Conclusion:-

*The p-value is greater than 0.05, so we fail to reject the null hypothesis. This indicates that there is no statistically significant difference in bike rentals between working and non-working days.

*This suggests that bike usage is not heavily dependent on working day status and may be influenced by other factors such as weather and season.

Observation:-

This indicates that Yulu bikes are used for both commuting and leisure, not strictly dependent on working days.

Summary:-

Although minor differences may be observed visually, statistical testing confirms whether those differences are meaningful. In this case, working day does not appear to be a strong driver of demand compared to environmental factors.

ANOVA(analysis of variance)

To check whether the mean bike rentals (count) differ across multiple groups like Weather conditions, Seasons.

a)Weather has 4 categories.

b)Season has 4 categories.

PART 1 (Weather vs Count)

Hypothesis:-

H_0 :Mean rentals are same across all weather conditions.

H_1 :At least one weather category has different mean rentals.

```
from scipy.stats import f_oneway

weather_groups = [df_bike_sharing[df_bike_sharing['weather'] == i]
['count'] for i in df_bike_sharing['weather'].cat.categories]
```

Data is not perfectly normal, but large sample size justifies ANOVA.

```
from scipy.stats import levene

levene(*weather_groups)

LeveneResult(statistic=np.float64(54.85106195954556),
pvalue=np.float64(3.504937946833238e-35))
```

If, $p > 0.05$ then it has equal variance, Although, the equal variance assumption is violated ($p < 0.05$), ANOVA can still be applied due to the large sample size, but results should be interpreted with caution.

Assumption:-

Although normality assumption is not strictly met, the large sample size ensures robustness of ANOVA results.

Anova test

```
f_oneway(*weather_groups)
F_onewayResult(statistic=np.float64(65.53024112793265),
pvalue=np.float64(5.482069475935669e-42))
```

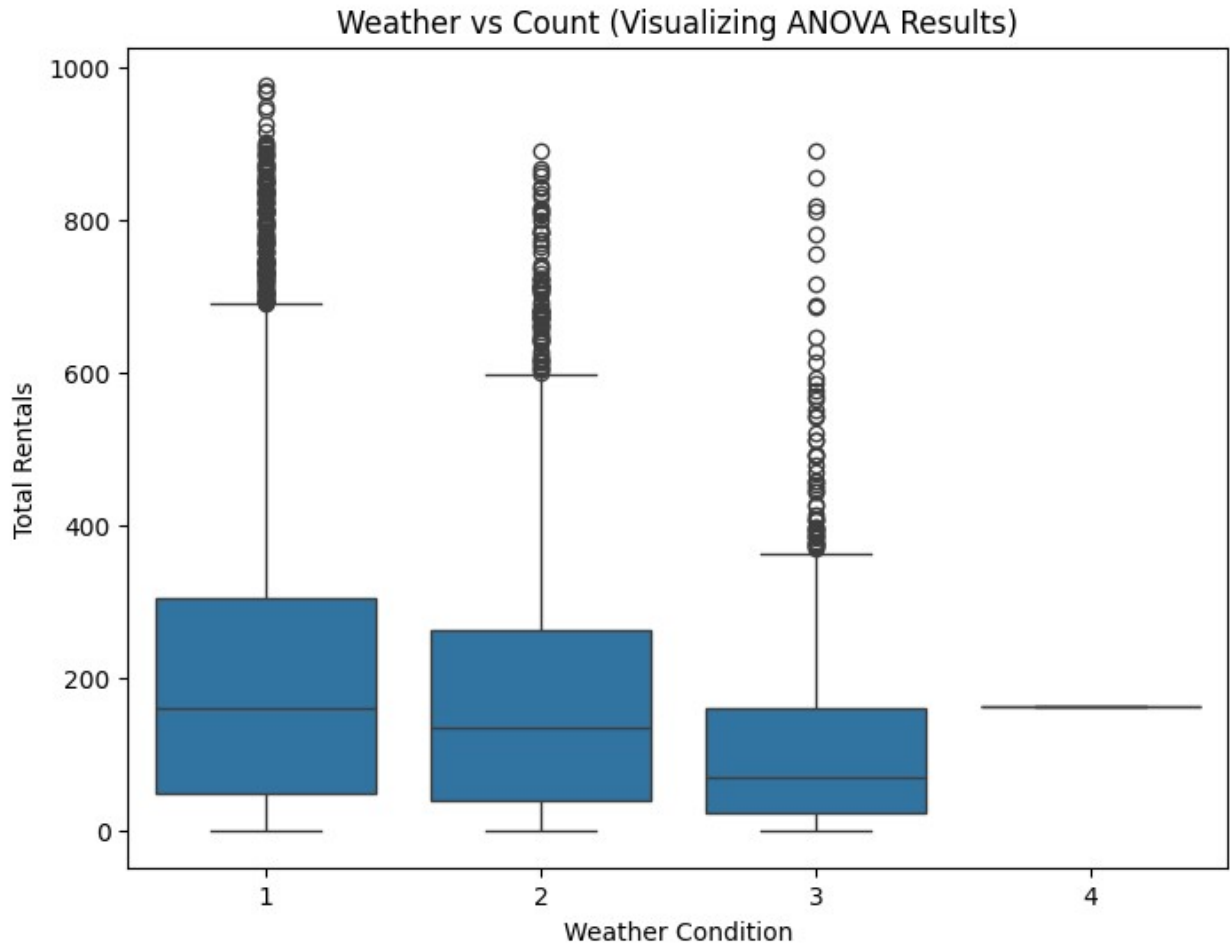
The p-value is less than 0.05, so we reject the null hypothesis.

summary:-

This indicates that bike rentals differ significantly across weather conditions.

```
plt.figure(figsize=(8, 6))
sns.boxplot(x='weather', y='count', data=df_bike_sharing)
plt.title('Weather vs Count (Visualizing ANOVA Results)')
plt.xlabel('Weather Condition')
plt.ylabel('Total Rentals')

plt.show()
```



The ANOVA results are consistent with the boxplots observed in EDA, where clear differences in distributions across categories were visible.

Observations:-

- a) Clear Differences in Medians: The median rental counts (the horizontal lines inside each box) are distinctly different
- b) Varying Distributions: The overall distribution of rental counts, represented by the box and whiskers, changes significantly.
- c) Impact of Adverse Conditions: As weather conditions worsen (moving from 1 to 3), the median and overall distribution of rentals tend to decrease, highlighting the negative impact of poor weather on bike usage.

Post-Hoc (tukey test)

```
from statsmodels.stats.multicomp import pairwise_tukeyhsd

tukey_weather = pairwise_tukeyhsd(
    endog=df_bike_sharing['count'],
    groups=df_bike_sharing['weather'],
```



```

    alpha=0.05
)
print(tukey_weather)

```

Multiple Comparison of Means - Tukey HSD, FWER=0.05						
group1	group2	meandiff	p-adj	lower	upper	reject
1	2	-26.2813	0.0	-36.5135	-16.049	True
1	3	-86.3905	0.0	-103.0452	-69.7357	True
1	4	-41.2368	0.9957	-502.6221	420.1485	False
2	3	-60.1092	0.0	-78.0783	-42.1401	True
2	4	-14.9555	0.9998	-476.3902	446.4791	False
3	4	45.1537	0.9944	-416.468	506.7754	False

Key insights (Tukey test) :-

Rentals in clear weather (weather 1) are significantly higher than misty and rainy conditions, indicating that worsening weather consistently reduces demand.

Tukey HSD results confirm that rentals significantly decrease as weather conditions worsen, with the largest drop observed between clear and rainy conditions, reinforcing the strong negative impact of adverse weather.

Observation:-

The statistical results align with patterns observed during EDA, reinforcing the reliability of findings.

Business insight:-

Weather has a strong negative impact on demand under poor conditions, suggesting that Yulu should adjust pricing or availability based on weather forecasts.

PART 2 (Season vs Count)

Hypothesis

H_0 : Mean rentals are same across all seasons.

H_1 : At least one season differs.

```

season_groups = [df_bike_sharing[df_bike_sharing['season'] == i]
['count'] for i in df_bike_sharing['season'].cat.categories]

```

Anova

```
f_oneway(*season_groups)
```

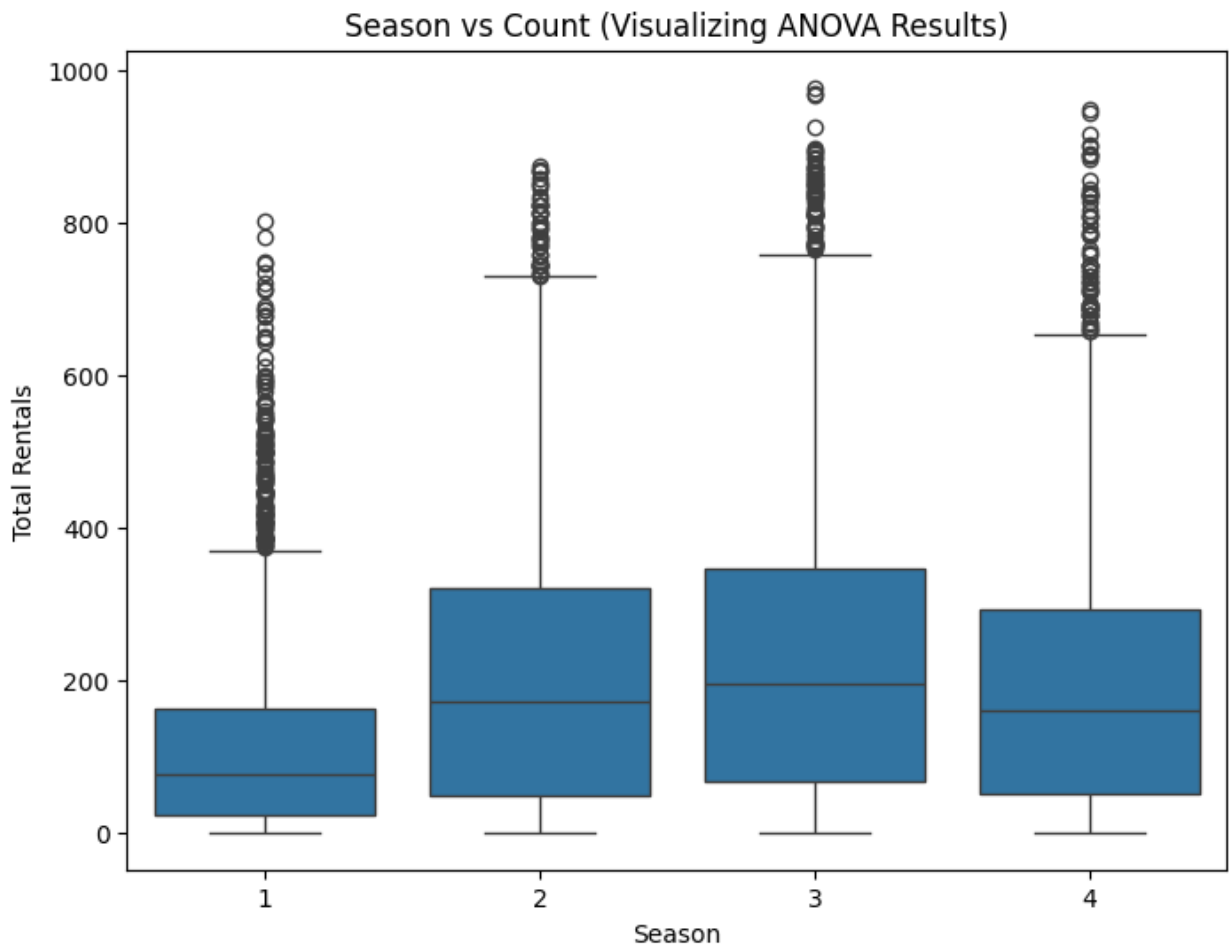
```
F_onewayResult(statistic=np.float64(236.94671081032098),  
pvalue=np.float64(6.164843386499654e-149))
```

The p-value is less than 0.05, so we reject the null hypothesis.

Observation:-

The extremely low p-value indicates strong evidence against the null hypothesis, confirming that season has a major impact on bike rental demand

```
plt.figure(figsize=(8, 6))  
sns.boxplot(x='season', y='count', data=df_bike_sharing)  
plt.title('Season vs Count (Visualizing ANOVA Results)')  
plt.xlabel('Season')  
plt.ylabel('Total Rentals')  
plt.show()
```



Key insights:-

a) Clear Differences in Medians: The median rental counts (the line inside each box) are distinctly different across the four seasons.

b)Varying Distributions: The overall distribution (the box and whiskers) of rental counts changes significantly with each season.

c)Higher Averages in Warmer Seasons: The box plots for seasons 2 and 3 are positioned higher on the 'Total Rentals'.

Conclusion :-

*This indicates that bike rentals differ significantly across seasons. Moreover, higher rentals are observed in summer and fall, while winter demonstrates lower demand, indicating strong seasonal influence on bike usage.

*These insights can be integrated into predictive models for demand forecasting and real-time operational optimization.

Mean comparison

```
df_bike_sharing.groupby('season')['count'].mean()
```

```
/tmp/ipykernel_17165/2901253952.py:1: FutureWarning: The default of observed=False is deprecated and will be changed to True in a future version of pandas. Pass observed=False to retain current behavior or observed=True to adopt the future default and silence this warning.
```

```
df_bike_sharing.groupby('season')['count'].mean()
```

```
season
```

```
1    116.343261
```

```
2    215.251372
```

```
3    234.417124
```

```
4    198.988296
```

```
Name: count, dtype: float64
```

```
df_bike_sharing.groupby('weather')['count'].mean()
```

```
/tmp/ipykernel_17165/3851374608.py:1: FutureWarning: The default of observed=False is deprecated and will be changed to True in a future version of pandas. Pass observed=False to retain current behavior or observed=True to adopt the future default and silence this warning.
```

```
df_bike_sharing.groupby('weather')['count'].mean()
```

```
weather
```

```
1    205.236791
```

```
2    178.955540
```

```
3    118.846333
```

```
4    164.000000
```

```
Name: count, dtype: float64
```

Observation:-

*The mean rentals vary significantly across weather conditions, with higher rentals in clear weather and lower rentals in adverse conditions.

*Rentals are higher in summer/fall and lower in winter.

Business insight:-

Seasonal trends indicate predictable demand cycles, which can be leveraged for fleet planning and marketing strategies.

Post-hoc(Tukey test)

```
from statsmodels.stats.multicomp import pairwise_tukeyhsd

tukey_season = pairwise_tukeyhsd(
    endog=df_bike_sharing['count'],
    groups=df_bike_sharing['season'],
    alpha=0.05
)

print(tukey_season)
```

```
Multiple Comparison of Means - Tukey HSD, FWER=0.05
=====
group1 group2 meandiff p-adj    lower    upper    reject
-----
    1     2   98.9081    0.0   86.6544  111.1618    True
    1     3  118.0739    0.0  105.8202  130.3276    True
    1     4   82.645    0.0   70.3924   94.8976    True
    2     3   19.1658  0.0003    6.9653   31.3662    True
    2     4  -16.2631  0.0034  -28.4624  -4.0637    True
    3     4  -35.4288    0.0  -47.6282  -23.2295    True
-----
```

All seasonal pairs indicates significant differences, with peak demand in summer and fall, highlighting strong seasonal demand cycles.

Post-Hoc (tukey test):-

Post-hoc analysis using Tukey HSD reveals which specific groups differ significantly. For example, rentals in clear weather differ significantly from rainy conditions, confirming that adverse weather reduces demand.

Conclusion:-

*ANOVA results confirm that both weather and season significantly impact bike rentals. Weather has a strong negative effect under poor conditions, while seasonality drives demand patterns throughout the year.

*The combination of EDA and statistical testing ensures both exploratory understanding and statistical validation of results.

Chi-square test

Weather has 4 categories

Season has 4 categories

Hypothesis:-

H_0 (Null): Weather is independent of season.

H_1 (Alternate): Weather is dependent on season.

```
contingency_table = pd.crosstab(df_bike_sharing['season'],
df_bike_sharing['weather'])
contingency_table

{"summary":{"\n  \"name\": \"contingency_table\",\n  \"rows\": 4,\n  \"fields\": [\n    {\n      \"column\": \"season\",\n      \"properties\": {\n        \"dtype\": \"category\",\n        \"num_unique_values\": 4,\n        \"samples\": [\n          2,\n          4,\n          1\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\",\n        \"properties\": {\n          \"dtype\": \"number\",\n          \"std\": 96,\n          \"min\": 1702,\n          \"max\": 1930,\n          \"num_unique_values\": 4,\n          \"samples\": [\n            1801,\n            1702,\n            1759\n          ],\n          \"semantic_type\": \"\",\n          \"description\": \"\",\n          \"properties\": {\n            \"dtype\": \"number\",\n            \"std\": 82,\n            \"min\": 604,\n            \"max\": 807,\n            \"num_unique_values\": 4,\n            \"samples\": [\n              708,\n              807,\n              715\n            ],\n            \"semantic_type\": \"\",\n            \"description\": \"\",\n            \"properties\": {\n              \"dtype\": \"number\",\n              \"std\": 12,\n              \"min\": 199,\n              \"max\": 225,\n              \"num_unique_values\": 4,\n              \"samples\": [\n                224,\n                225,\n                211\n              ],\n              \"semantic_type\": \"\",\n              \"description\": \"\",\n              \"properties\": {\n                \"dtype\": \"number\",\n                \"std\": 0,\n                \"min\": 0,\n                \"max\": 1,\n                \"num_unique_values\": 2,\n                \"samples\": [\n                  0,\n                  1\n                ],\n                \"semantic_type\": \"\",\n                \"description\": \"\"\n              }\n            }\n          }\n        }\n      }\n    }\n  ],\n  \"type\": \"dataframe\", \"variable_name\": \"contingency_table\"}
```

Chi-Square test

```
from scipy.stats import chi2_contingency
```

```
chi2, p, dof, expected = chi2_contingency(contingency_table)
```

```
print("Chi2:", chi2)
```

```
print("p-value:", p)
```

```
Chi2: 49.158655596893624
```

```
p-value: 1.549925073686492e-07
```

$\alpha = 0.05$ So, $p < 0.05$ then Reject H_0

The observed frequencies deviate noticeably from expected frequencies under independence, confirming that season and weather are statistically associated.

Key insight :-

*Certain weather conditions occur more frequently in specific seasons, confirming a relationship between season and weather.

*The observed frequencies differ from expected frequencies under independence, further supporting the relationship between season and weather.

Assumption :-

The expected frequencies are sufficiently large, satisfying the assumptions of the Chi-square test.

Business insight :-

Since weather depends on season and weather significantly affects demand, seasonal planning becomes critical for optimizing bike availability and operations.

Chi-square test:-

A Chi-square test of independence was performed to examine the relationship between season and weather. The p-value obtained was less than 0.05, leading to rejection of the null hypothesis. This indicates that weather conditions are significantly dependent on season. This relationship is expected as different seasons exhibit distinct weather patterns.

#6.SUMMARY

Summary :-

*Weather and season are the most influential factors affecting bike rental demand.

*Rentals decrease significantly under poor weather conditions.

*Seasonal trends depicts higher demand in moderate climates and lower in extreme conditions.

*Working day has minimal impact, indicating mixed usage behavior (commuting + leisure).

*Temperature positively influences demand, while extreme environmental conditions reduce usage.

*Mean rentals in clear weather are significantly higher (approximately 205) than during rainy conditions (approximately 118), representing a substantial 40–45% drop in demand."

Based on statistical analysis, the key drivers of demand can be ranked as:

1.Weather: Strong negative impact under adverse conditions.

2.Season: Predictable demand cycles across the year

3.Temperature: Positive influence within optimal range

4.Working Day: Minimal influence

#7.RECOMMENDATIONS

Business recommendations :-

*Increase bike availability dynamically during favorable weather and peak seasons to maximize utilization.

*Introduce dynamic pricing or discounts during poor weather to sustain demand.

*Optimize fleet distribution based on seasonal demand trends.

*Use weather forecasts to plan operations proactively.

*Promote usage during weekends and leisure periods.

Business decisions :-

These insights can help Yulu improve operational efficiency, increase utilization, and optimize revenue through data-driven decision-making.

#8.LIMITATIONS

Limitations :-

*The dataset does not include factors such as pricing, location, or user demographics.

*Some assumptions (like normality) are not strictly met but are handled using large sample size.

*External factors like fuel prices or public transport availability are not considered.

*The variable count is derived from casual and registered, so care must be taken in modeling scenarios to avoid redundancy or data leakage.